Name: ____________________

PERCENTAGES OF AMOUNTS, (PERCENTAGE INCREASE AND DECREASE)

Date: ______________

LO: I can find a percentage increase or decrease and express a change as a percentage.

Percentage change

A percentage change tells you how much something has gone up or down compared to the original amount, expressed as a percentage.

Formula: Percentage change = (change ÷ original) × 100%

Quick examples

●	20 to 25: change = 5, % = $\frac{5}{20}$ × 100 = 25%	increase
●	80 to 60: change = 20, % = $\frac{20}{80}$ × 100 = 25%	decrease
●	Always divide by the ORIGINAL amount	key rule
●	20 to 25: % change = $\frac{5}{25}$ × 100 = 20%	divided by new, not original

Key idea

Percentage change = (difference ÷ original) × 100. Always divide by the ORIGINAL value.

$$\% \text{ change} = \frac{\text{change}}{\text{original}} \times 100$$

Divide by the starting value, not the new value

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – percentage increase

A price goes from £40 to £52. Find the percentage increase.

Change = 52 - 40 = 12

% change = 12 ÷ 40 × 100

= 30% increase

= 30% increase

Divide the change by the original amount.

Example 2 – percentage decrease

A weight drops from 80 kg to 68 kg. Find the percentage decrease.

Change = 80 - 68 = 12

% change = 12 ÷ 80 × 100

= 15% decrease

= 15% decrease

Same formula, but the answer is a decrease.

Example 3 – with decimals

A bill goes from £120 to £135. Find the percentage change.

Change = 135 - 120 = 15

% change = 15 ÷ 120 × 100

= 12.5% increase

= 12.5% increase

The answer can be a decimal percentage.

Example 4 – comparing changes

Shop A: £50 to £60. Shop B: £80 to £92. Which has the bigger % increase?

Shop A: $\frac{10}{50}$ × 100 = 20%

Shop B: $\frac{12}{80}$ × 100 = 15%

Shop A has the bigger percentage increase

Shop A = 20%, Shop B = 15%

Compare percentage changes, not absolute changes.

YOUR TURN – PRACTICE (deliberate practice)

1. Finding percentage increase
Find the percentage increase.

a) 20 to 25

b) 50 to 65

c) 80 to 100

d) 200 to 230

2. Finding percentage decrease
Find the percentage decrease.

a) 40 to 30

b) 60 to 45

c) 120 to 96

d) 500 to 350

3. Percentage change with decimals
Find the percentage change (to 1 d.p. if needed).

a) 30 to 38

b) 75 to 60

c) 110 to 143

d) 64 to 48

4. Real-world percentage change
Calculate the percentage change.

a) Rent goes from £600 to £660

b) Temperature drops from 20°C to 15°C

c) Population grows from 4000 to 4500

d) Price falls from £90 to £72

5. Mixed percentage change
Identify increase or decrease and find the percentage.

a) 45 to 54 **b) 160 to 120** **c) £35 to £42** **d) 250 g to 200 g** **e) £1200 to £1350**

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) Percentage change = change ÷ original × 100 ☐	b) From 20 to 25 is a 20% increase ☐	c) Always divide by the original (starting) value ☐	d) A change from 80 to 60 and 60 to 80 are the same percentage ☐	e) From 50 to 65 is 30% increase ($\frac{15}{50}$ × 100) ☐
Correct answer: __________	Correct answer: __________	Correct answer: __________	Correct answer: __________	Correct answer: __________
Reason: ________________	Reason: ________________	Reason: ________________	Reason:	Reason: ________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A shop buys jumpers for **£24** and sells them for **£36**. Find the percentage profit. If they then put a **25% off** sticker on unsold jumpers, do they still make a profit? Explain.

Expression: ____________________________

Simplified: ____________________________

b) In January, a town has **8000 residents**. By December, it has **8640**. Find the percentage increase. If the same percentage increase happens next year, predict the population for December of the following year.

Expression: ____________________________

Simplified: ____________________________

TOP TIPS
Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK
Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!